LAB PYTHON PROGRAMS
CSA1718-ARTIFICIAL INTELLIGENCE

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2. **Write a Python Program to Solve the 8-Queen Problem**

PROGRAM:

```
N = 8
board = [-1] * N

def safe(row, col):
 for i in range(row):
 if (board[i] == col or
 abs(board[i] - col) == abs(i - row)):
 return False
 return True

def solve(row):
 if row == N:
 return True

 for col in range(N):
 if safe(row, col):
 board[row] = col
 if solve(row + 1):
 return True
 board[row] = -1

 return False

if solve(0):
 for row in range(N):
 for col in range(N):
 print("Q" if board[row] == col else ".", end=" ")
 print()
else:
```

```
print("No solution")
```

OUTPUT:

